Overparameterized Nonnegative Tensor Factorization for Hyperspectral Unmixing

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***Abstract*—Hyperspectral unmixing decomposes each pixel into constituent spectral signatures (endmembers) and estimates their corresponding proportions (abundances) to address spectral mixing caused by limited spatial resolution. While low-rank tensor factorization can exploit spatial–spectral structures, its shallow formulation and the challenge of selecting the tensor rank limit its ability to characterize complex, low-rank relationships in abundance maps. Recent findings reveal that gradient-based optimization induces an implicit regularization, enabling overparameterized models where parameters vastly outnumber training samples to learn powerful representations that often generalize well. Motivated by this, we propose an overparameterized nonnegative tensor factorization (ONTF) for hyperspectral unmixing. Under the linear mixing model (LMM) framework, ONTF embeds abundance estimation into a stack of nested nonlinear factorizations, forming an implicit low-rank encoder. The endmembers act as an explicit linear decoder for reconstruction. Both encoder and decoder are jointly optimized via gradient descent. Benefiting from implicit regularization, ONTF gains an enhanced capacity to recover low-rank abundance structures with complex nonlinear dependencies while preserving physical interpretability. Experiments on synthetic and real-world hyperspectral data demonstrate that ONTF consistently outperforms state-of-the-art unmixing methods.**

***Index Terms*—Deep learning, hyperspectral unmixing, implicit regularization, low-rank tensor factorization.**

I. Introduction

HYPERSPECTRAL imaging (HSI) records a continuous spectral curve for each pixel, forming a 3-D data cube that contains rich spectral information alongside spatial information. However, this high spectral resolution is typically obtained at the expense of spatial resolution, resulting in mixed pixels that contain signals from multiple materials [1]. Hyperspectral unmixing aims to decompose each pixel into a combination of constituent materials, known as endmembers, and their corresponding proportions or abundances [2]. Thereby, this process reveals the true spatial distribution of materials and significantly enhances the utility of HSIs.

The unmixing problem can be considered as an ill-posed inverse problem. To perform unmixing effectively, it is essential to adopt a physical model that describes the mixing process of endmembers and their corresponding fractional abundances. The linear mixing model (LMM) assumes that incoming light interacts with only a single material before reaching the sensor [3]. In contrast, nonlinear mixture models assume that the light interacts with multiple materials [4], [5]. Due to its mathematical tractability and computational efficiency, the LMM remains the most widely adopted model in practice [6], [7].

When endmembers are assumed to originate from a preconstructed spectral library, the problem is formulated as semi-supervised unmixing. Here, sparse regression methods are employed to select a small set of representative endmembers from the large library [8], [9], [10], [11]. To further exploit spatial contextual information, spatial regularization—such as total variation (TV)—is often incorporated to enforce spatial smoothness and material consistency [12], [13]. When limited prior knowledge of endmembers is available, hyperspectral unmixing is typically formulated as an unsupervised or blind source separation problem. In this case, the problem can be reformulated as a low-rank representation of the high-dimensional HSI, where the endmembers act as a low-rank dictionary and the abundances serve as the representation coefficients. Nonnegative matrix factorization (NMF) is widely adopted in this setting. In NMF, the 3-D HSI cube is first reshaped into a 2-D matrix and then decomposed into the product of two nonnegative matrices. However, the objective function of NMF is nonconvex, often leading to numerous local optima. To mitigate this issue and constrain the solution space, various regularizers based on prior knowledge such as minimum-volume [14], sparsity [15], [16], and spatial smoothness [17], [18] have been introduced to improve the stability and robustness of the unmixing process.

However, representing 3-D HSIs as 2-D matrices inevitably destroys the intrinsic spatial structure and joint spatial–spectral correlations. In contrast, 3-D HSIs can be naturally modeled as third-order tensors. Low-rank tensor decompositions effectively capture these inherent multidimensional dependencies, thereby offering improved representational capability for hyperspectral unmixing [19], [20], [21]. Building on this, Qian et al. [22] established a connection between nonnegative tensor factorization (NTF) and the LMM and introduced matrix-vector NTF (MV-NTF) for unmixing. MV-NTF decomposes an HSI into multiple component tensors, each represented as the outer product of a low-rank

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abundance matrix and a 1-D endmember vector. Leveraging this inherent low-rank representation in both spatial and spectral domains, MV-NTF achieves superior unmixing performance. Nevertheless, similar to NMF, the objective function of NTF remains nonconvex, often leading to numerous local optima. To address this, various regularization strategies such as TV [23] and low-rank constraints [24], [25] have been incorporated to promote spatial coherence and reduce solution ambiguity. However, this strategy introduces additional complexity, as it requires careful tuning of multiple regularization parameters.

Despite these advances, learning effective representations remains a fundamental challenge in hyperspectral unmixing. With the advent of deep learning and its strong nonlinear modeling capability, deep architectures have demonstrated superior representational power for this task [26], [27], [28]. Among them, autoencoders (AEs) have become a popular framework for learning unsupervised representations, showing strong performance in hyperspectral unmixing [29], [30]. In these frameworks, the encoder maps the HSI into a low-dimensional latent space to estimate the abundance maps, while the decoder reconstructs the original spectral data from these abundances using the corresponding endmembers. To better capture the spatial structures of abundances and prevent the AE from learning trivial representations, various attention mechanisms, such as spatial attention [31], channel attention [32], and Transformer blocks [3], have been incorporated. Additionally, introducing explicit regularization terms into the objective function is a common strategy to enhance unmixing performance. For example, Qu and Qi [33] incorporated an L_{21}-norm to suppress redundant endmembers and reduce reconstruction error. The unmixing process can also be interpreted as a sampling procedure from a posterior distribution. Building on this perspective, Deng et al. [34] and Wu et al. [35] integrated diffusion models (DMs) into linear and nonlinear unmixing frameworks, respectively.

The main limitation of AE-based hyperspectral unmixing lies in its limited interpretability and weak incorporation of prior physical knowledge. To address this, model-based deep learning has emerged as a hybrid paradigm that combines the interpretability and theoretical grounding of model-driven approaches with the powerful representation learning capability of data-driven networks [36], [37]. A common strategy is to unfold the iterative steps of a traditional numerical optimization algorithm into a sequence of neural network layers, forming a deep unrolled architecture. For example, Qian et al. [38] unfolded the optimization process of a sparse mixture model into a learnable network named MNN-BU. This network follows an autoencoder-like structure where the encoder performs sparse abundance estimation and the decoder represents the endmembers. Xiong et al. [39] further extended MNN-BU by jointly unfolding both endmember and abundance estimation within a unified network. However, deep unfolding-based methods still face several challenges. First, the optimization process is constrained to a finite, often small, number of unrolled iterations, as deeper networks are difficult to train effectively, limiting their representational capacity [40], [41]. Second, most existing methods reshape the 3-D HSI into a 2-D matrix for processing, thereby destroying the intrinsic joint spatial–spectral structure of the data. This leads to a central question motivating our work. Is there an alternative model-based deep learning approach that can inherit the interpretability of physical models while fully preserving and exploiting the 3-D spatial–spectral structure of HSIs?

To address this objective, we revisit the formulation of MV-NTF. In essence, MV-NTF can be interpreted as an autoencoder, where the explicit low-rank factorization of the abundance maps learns a low-dimensional spatial representation. Recent theoretical studies have shown that optimizing an unregularized model via gradient descent can implicitly induce a low-rank bias, enhancing generalization even in overparameterized settings [42], [43], [44]. This phenomenon is known as implicit regularization [42], [43], [44]. Implicit regularization describes the tendency of certain optimization algorithms to converge toward solutions with favorable generalization properties, even without explicit regularization terms in the loss function. Viewing matrix factorization as a two-layer linear network, it has been shown theoretically that gradient descent acts as an implicit nuclear norm regularizer, steering solutions toward low-rank matrices. Extending this insight, Arora et al. [45] demonstrated that increasing network depth in deep matrix factorization strengthens this implicit bias toward lower rank representations. Razin et al. [46] further generalized this principle to tensor CANDECOMP/PARAFAC (CP) decomposition, proving that gradient descent exhibits an implicit bias toward low tensor rank. More recently, this phenomenon has also been established for Tucker decomposition [47], [48], where gradient-based optimization naturally promotes solutions with low multilinear rank.

Motivated by these insights into implicit regularization, we investigate whether this phenomenon can be integrated into the MV-NTF framework to better capture the low-rank spatial structure of abundances while enhancing representational capacity. To this end, we propose an overparameterized nonnegative tensor factorization (ONTF) model for hyperspectral unmixing. MV-NTF explicitly constrains the rank of abundance maps by limiting the shared dimensionality between two thin matrices. In contrast, ONTF introduces a series of nonlinear layers that form a deep matrix factorization of the abundance maps, enabling the model to implicitly learn low-rank spatial structures. In this way, the tensor factorization contains multiple encoders, each responsible for one abundance component. The outputs of all encoders are passed through a softmax layer to satisfy the abundance sum-to-one constraint (ASC) and are jointly optimized with the endmember matrix, which serves as the decoder, via standard gradient backpropagation. By integrating implicit regularization into the MV-NTF framework, ONTF moves beyond shallow linear representations and can more effectively capture complex, low-rank spatial–spectral dependencies in HSIs. Crucially, since the estimated abundances remain linearly combined with the endmembers, ONTF retains adherence to the LMM, preserving physical interpretability. Consequently, the proposed ONTF framework achieves state-of-the-art unmixing

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performance on both synthetic and real-world hyperspectral datasets.

We acknowledge several prior studies that have explored deep matrix or tensor factorizations for hyperspectral unmixing. For instance, Li et al. [49] and Feng et al. [50] extended NMF into an autoencoder-like framework where the abundance matrix is represented through multiple layers. However, the specific properties learned by such a multilayer encoder remain unclear. Zheng et al. [51] extended the MV-NTF framework in a hierarchical manner with additional sparse constraints, yet the physical interpretation of the hierarchical endmember decomposition and its connection to the LMM remains ambiguous. Our proposed ONTF method differs from these approaches in two fundamental aspects. First, our ONTF maintains a clear link to the LMM. The hierarchical, nonlinear representation is embedded within each abundance component encoder, explicitly pursuing low-rank abundance representations while keeping the spectral mixing process linear and interpretable. Second, unlike prior works that rely on complex, explicitly designed regularizers (e.g., sparsity in [49], [50], and [51]), our approach leverages the implicit regularization inherent in gradient-based optimization of overparameterized models. Benefiting from recent theoretical insights into deep matrix and tensor factorizations, the proposed ONTF framework more effectively exploits the intrinsic spatial–spectral low-rank structure of hyperspectral images (HSIs), leading to enhanced unmixing performance.

The remainder of this article is organized as follows. Section II introduces the background of tensor-based unmixing. Section III introduces the details of the proposed ONTF model. Section IV presents the quantitative and visual experiments. Finally, Section V concludes this article.

II. Background

This section describes the linear mixture model and low-rank tensor factorization-based unmixing.

A. Linear Mixture Model

Let $\mathbf{X}$ be an HSI with $N = I \times J$ pixels and K-bands, where I and J represent the width and height, respectively. Under the linear mixture model, the observed pixel spectrum is assumed to be a linear combination of constituent endmember spectra, i.e.,

$$\mathbf{X} = \mathbf{MS} + \mathbf{N}. \tag{1}$$

Here, $\mathbf{M} = [\mathbf{m}_1, \ldots, \mathbf{m}_R] \in \mathbb{R}^{K\times R}$ denotes the endmember matrix, $\mathbf{S} \in \mathbb{R}^{R\times N}$ represents the abundance matrix (with each column specifying the fractional proportion of each endmember for a given pixel), and $\mathbf{N} \in \mathbb{R}^{K\times N}$ accounts for noise and model errors.

To ensure physically meaningful unmixing, two constraints are usually imposed on the abundances. The abundance nonnegativity constraint (ANC) enforces that all abundance values must be nonnegative. The ASC ensures that the fractions associated with all endmembers for each pixel sum to one. Mathematically, the ASC and ANC can be represented by the following equation:

$$\mathbf{S} \geq 0$$

$$\mathbf{1}_R^\top \mathbf{S} = \mathbf{1}_N^\top. \tag{2}$$

B. Matrix-Vector Nonnegative Tensor Factorization

Traditionally, the hyperspectral unmixing is performed under the matrix notation where the 3-D HSI is unfolded into a 2-D matrix. This limitation results in the spatial relationship loss. Under the tensor notation, an HSI can be naturally represented as a third-order tensor $\mathcal{X} \in \mathbb{R}^{I\times J\times K}$, where the first two modes correspond to the spatial dimensions and the third mode corresponds to the spectral dimension. In this case, the LMM can be equivalently expressed as follows:

$$\mathcal{X} = \sum_{r=1}^{R} \mathbf{S}_r \circ \mathbf{m}_r + \mathcal{N} \tag{3}$$

where $\mathbf{S}_r \in \mathbb{R}^{I\times J}$ is the abundance map associated with the rth endmember $\mathbf{m}_r \in \mathbb{R}^{K\times 1}$ and $\mathcal{N} \in \mathbb{R}^{I\times J\times K}$ is the noise tensor. Here, $\circ$ denotes the outer product between two tensors $\mathcal{A}_{i_1 i_2 \cdots i_p}$ and $\mathcal{B}_{j_1 j_2 \cdots j_q}$, defined as follows:

$$\mathcal{C}_{i_1 i_2 \cdots i_p j_1 j_2 \cdots j_q} = \mathcal{A}_{i_1 i_2 \cdots i_p} \cdot \mathcal{B}_{j_1 j_2 \cdots j_q}. \tag{4}$$

MV-NTF further decomposes each abundance map $\mathbf{S}_r$ into the product of two thin matrices, i.e.,

$$\mathbf{S}_r = \mathbf{U}_r \mathbf{V}_r^\top \tag{5}$$

where $\mathbf{U}_r \in \mathbb{R}^{I\times L_r}$ and $\mathbf{V}_r \in \mathbb{R}^{J\times L_r}$ with $L_r < min(I, J)$. Under this formulation, the unmixing model becomes

$$\mathcal{X} = \sum_{r=1}^{R} \mathbf{U}_r \mathbf{V}_r^\top \circ \mathbf{m}_r + \mathcal{N}. \tag{6}$$

As L_r is much smaller than the spatial dimensions, this factorization imposes a low-rank structure on the abundance maps. As a result, MV-NTF captures spatial correlations more effectively and improves robustness against noise.

However, MV-NTF also has notable limitations. First, the hyperparameter L_r is difficult to determine in practice. Though Qian et al. [22] experimentally set $L_r = 2/3 \min(I, J)$ for $r = 1, 2, \ldots, R$, this choice does not generalize well across different datasets for scene complexities. On the other hand, MV-NTF remains essentially a shallow model, where the representation capacity is limited by the fixed low-rank constraint. As a result, its ability to fully exploit the complex spatial structures embedded in abundance maps is inherently constrained.

III. ONTF for Hyperspectral Unmixing

In this section, we introduce the details of our ONTF.

A. Overparameterized Nonnegative Tensor Factorization

As the depth of neural networks increases, the number of learnable parameters can rapidly exceed the number of available training samples, placing the model in a heavily overparameterized regime. In the face of overparameterization, the model is more likely prone to overfitting. Surprisingly, such models often still generalize well. This phenomenon has been partially explained by the theory of implicit regularization, which suggests that gradient-based optimization

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inherently converges to low-dimensional solutions. Low-rank matrix factorization provides an effective test bed for studying implicit regularization. In this setting, the observed matrix $\mathbf{W}$ is parameterized as the product of two much smaller matrices. With the L_2 loss function, the objective is

$$\arg\min_{\mathbf{W}_1,\mathbf{W}_2} \frac{1}{2}\|\mathbf{W}-\mathbf{W}_2\mathbf{W}_1\|_F^2. \tag{7}$$

The solution is typically obtained by optimizing $\mathbf{W}_1$ and $\mathbf{W}_2$ using gradient descent. Gunasekar et al. [42] demonstrated that performing gradient descent with a small step size and an initialization close to zero implicitly biases the solution toward low-rank matrices. Remarkably, even when no explicit rank constraint is imposed and the factorization is overparameterized, i.e., $\mathbf{W}_1 = \mathbf{W}_2^T$, gradient descent converges to a minimal nuclear norm solution.

Subsequently, Arora et al. [45] extended these ideas to deep matrix factorization, in which the matrix is overparameterized as follows:

$$\mathbf{W} = \mathbf{W}_L\mathbf{W}_{L-1}\cdots\mathbf{W}_1 \tag{8}$$

where L denotes the depth. Empirically, increasing depth improves low-rank recovery performance under the framework of gradient descent-based optimization. Moreover, they also showed that when $L \geq 2$, the resulting implicit regularization can no longer be characterized as the minimization of any explicit matrix norm. This indicates that the implicit bias induced by gradient descent in deep factorizations is fundamentally richer and cannot be captured by classical norm-based formulations.

Building on the insights from implicit regularization in deep matrix factorizations, we aim to deepen the MV-NTF model to achieve improved unmixing performance. In hyperspectral unmixing, the decomposition must remain physically meaningful, requiring adherence to the linear spectral mixing model. Therefore, the introduced deep factorization should preserve a clear correspondence between the factorized matrices and the underlying endmembers and abundance maps, rather than producing latent factors that lack interpretability.

In order to fulfill the above goal, we keep the linear relationships between the abundance maps and endmembers so that our method still obeys the LMM. Instead, we deepen the representation of the abundances to achieve better low rankness so that the unmixing model has more capabilities to model the complex relationship within abundance maps and handle unexpected interruptions such as noise. Intuitively, the low-rank decomposition in (8) can be viewed as a two-layer neural network with linear activations. In the original MV-NTF, the spatial component is typically modeled as a low-rank matrix $\mathbf{S}_r$, decomposed into two thin factor matrices, $\mathbf{U}_r$ and $\mathbf{V}_r^\top$. To deepen this representation and exploit the implicit low-rank bias of deep networks, we apply the factorization from (8) to both $\mathbf{U}_r$ and $\mathbf{V}_r^\top$ simultaneously. Specifically, we expand the single-layer factors into multilayer products

$$\mathbf{U}_r \to \mathbf{U}_{r,L}\cdots\mathbf{U}_{r,1} \quad \text{and} \quad \mathbf{V}_r^\top \to \mathbf{V}_{r,1}^\top\cdots\mathbf{V}_{r,L}^\top.$$

Substituting these deep factors back into the spatial component $\mathbf{S}_r$ yields

$$\mathbf{S}_r = \underbrace{\mathbf{U}_{r,L}\mathbf{U}_{r,L-1}\cdots\mathbf{U}_{r,1}}_{\text{Deep } \mathbf{U}_r}\cdot\underbrace{\mathbf{V}_{r,1}^\top\cdots\mathbf{V}_{r,L-1}^\top\mathbf{V}_{r,L}^\top}_{\text{Deep } \mathbf{V}_r^\top} \tag{9}$$

where $\mathbf{U}_{r,l} \in \mathbb{R}^{d_l\times d_{l-1}}$ and $\mathbf{V}_{r,l} \in \mathbb{R}^{d_l\times d_{l-1}}$ for $l = 1,\cdots,L$. The quantity $2L$ is referred to as the depth of the factorization, and $\mathbf{U}_{r,l}$ and $\mathbf{V}_{r,l}$ are the factor matrices.

To further enhance the nonlinear representational power, nonlinear activations can be incorporated, yielding

$$\mathbf{S}_r = \mathbf{U}_{r,L}\sigma\left(\mathbf{U}_{r,L-1}\cdots\sigma\left(\mathbf{U}_{r,1}\mathbf{V}_{r,1}^\top\right)\cdots\mathbf{V}_{r,L-1}^\top\right)\mathbf{V}_{r,L}^\top \tag{10}$$

where $\sigma(\cdot)$ is the rectified linear unit activation function. It can be observed that we do not apply nonlinear activation functions to each factor $\mathbf{U}$ and $\mathbf{V}$, but rather to their product $\mathbf{U}_{r,l}\mathbf{V}_{r,l}^\top$. This design choice aims to preserve the strong interpretability of MV-NTF, where $\mathbf{U}_{r,l}\mathbf{V}_{r,l}^\top$ can be interpreted as a downsampled abundance map when $d_l \geq d_{l-1}$. By setting $d_l < \min(I,J)$ and through progressive nonlinear multiplication, the model is able to achieve a hierarchical multiscale low-rank representation of the data. Moreover, when the number of parameters

$$p = \sum_{l=1}^{L} 2d_l \times d_{l-1} \tag{11}$$

satisfies $p \geq I \times J$, we formulate an overparameterized representation of each abundance map.

In order to satisfy the ASC, we normalize $\mathbf{S}_r$ with a softmax function $\delta(\cdot)$, i.e.,

$$\delta(\mathbf{S}_r) = \frac{\exp^{\mathbf{S}_r}}{\sum_{r=1}^{R}\exp^{\mathbf{S}_r}}. \tag{12}$$

In this way, the model in (6) becomes

$$\mathcal{X} = \sum_{r=1}^{R}\delta(\mathbf{S}_r)\circ\mathbf{m}_r + \mathcal{N}. \tag{13}$$

In fact, the proposed ONTF can be interpreted as a neural network, as illustrated in Fig. 1. The abundance maps are overparameterized through a stack of nested low-rank matrix factorizations, which act as an implicit encoder, while the endmembers explicitly serve as the decoder. Such an overparameterized scheme offers two key advantages. First, unlike MV-NTF, which requires careful tuning of the rank parameter L_r to achieve the desired solution, the proposed approach can automatically learn low-rank solutions through gradient descent-based optimization. Second, in contrast to regularized MV-NTF formulations, the resulting problem is end-to-end, making it feasible to leverage efficient gradient-based optimization methods for optimization.

B. Gradient Descent-Based Optimization

To optimize the proposed nested nonlinear tensor factorization model in (13), we adopt a gradient descent framework. The objective function with mean-squared reconstruction loss is

$$\mathcal{L} = \frac{1}{2}\left\|\mathcal{X} - \sum_{r=1}^{R}\delta(\mathbf{S}_r)\circ\mathbf{m}_r\right\|_F^2. \tag{14}$$

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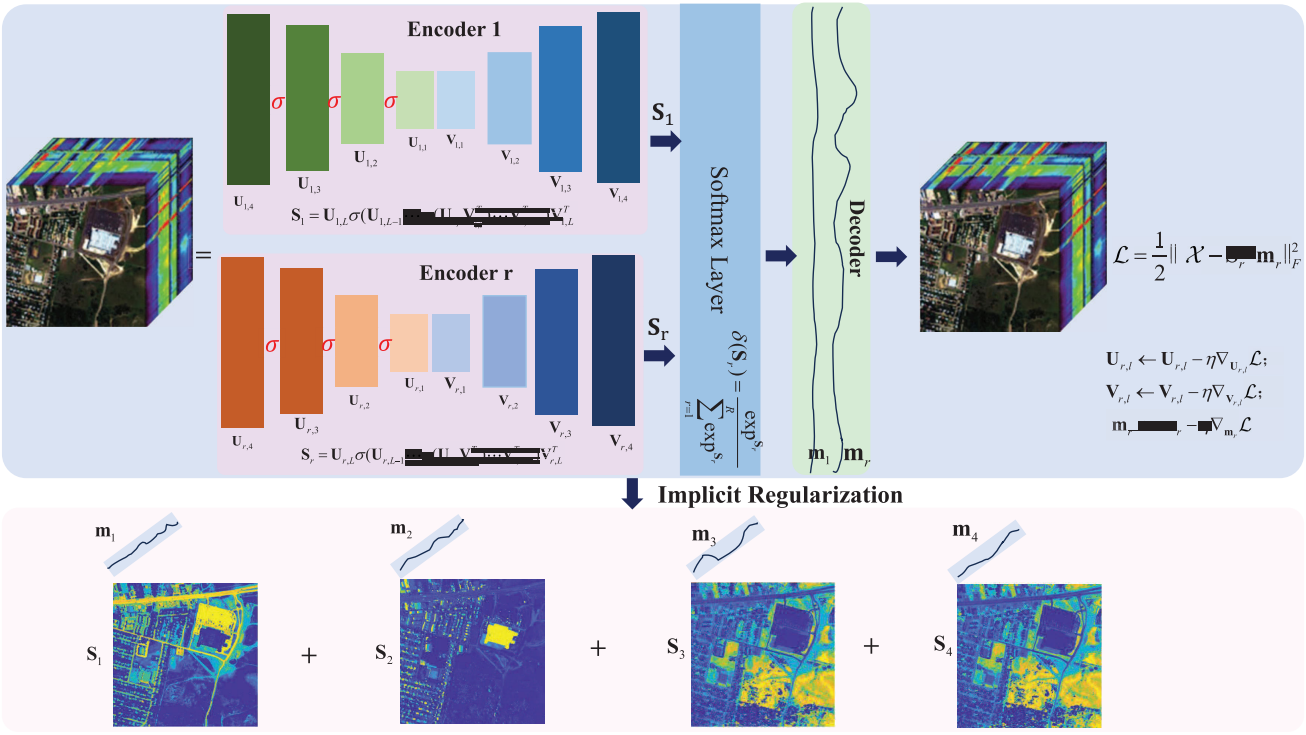


Fig. 1. Framework of the proposed ONTF unmixing method. The model takes an input HSI and learns a hierarchical overparameterized representation of the abundance maps. These maps are processed by a softmax layer to enforce the ASC and ANC constraints, followed by a decoder (initialized via the VCA algorithm) to yield the reconstructed HSI. The entire framework is trained via gradient descent, which induces implicit regularization, effectively promoting the estimation of low-rank structures.

For notational clarity, we denote the recursive nonlinear factorization of the abundance for the rth term as follows:

$$\begin{aligned}\mathbf{S}_{r,1} &= \mathbf{U}_{r,1}\mathbf{V}_{r,1}^{\top}\\ \mathbf{S}_{r,l+1} &= \mathbf{U}_{r,l+1}, \sigma\left(\mathbf{S}_{r,l}\right)\mathbf{V}_{r,l+1}^{\top}, \quad l = 1,\ldots,L-1.\end{aligned} \tag{15}$$

1) Forward Propagation: For each component r, the recursion in (15) is evaluated until $\mathbf{S}_{r,L}$ is obtained. Then, $\mathbf{S}_{r,L}$ is normalized using the softmax-based operator in (12), resulting in the abundance map $\mathbf{S}_r$. The output of the proposed ONTF model is finally computed as follows:

$$\widehat{\mathcal{X}} = \sum_{r=1}^{R} \mathbf{S}_r \circ \mathbf{m}_r. \tag{16}$$

2) Backward Propagation: Let $\mathbf{E} \in \mathbb{R}^{N\times K}$ be the matrix version of the residual $(\widehat{\mathcal{X}} - \mathcal{X})$, and the gradient of the loss with respect to $\text{vec}(\mathbf{S}_r)$ is

$$\frac{\partial \mathcal{L}}{\partial \text{vec}\left(\mathbf{S}_r\right)} = \mathbf{E}\mathbf{m}_r. \tag{17}$$

Define $\Delta_{r,k} = (\partial\mathcal{L})/(\partial\mathbf{S}_{r,l})$ to be the gradient at $l = L$. It is calculated by the following equation:

$$\text{vec}\left(\Delta_{r,N}\right) = \delta'\left(\text{vec}\left(\mathbf{S}_r\right)\right)\frac{\partial \mathcal{L}}{\partial \text{vec}\left(\mathbf{S}_r\right)} \tag{18}$$

where δ' is the derivative of the softmax function. For $l = L, L-1,\ldots,2$, the gradients are propagated backward as follows:

$$\mathbf{G}_{r,l-1} = \mathbf{U}_{r_l}^{\top}\Delta_{r,l}\mathbf{V}_{r,l} \tag{19}$$

$$\Delta^{r,l-1} = \mathbf{G}_{r,l-1} \circ \sigma'\left(\mathbf{S}_{r,l-1}\right) \tag{20}$$

where σ' denotes the elementwise derivative of the activation. For $k \geq 2$, where $\mathbf{A}_{r,l} = \mathbf{U}_{r_l}\mathbf{S}_{r,l-1}\mathbf{V}_{r_l}^{\top}$, the gradients for the factor matrices are

$$\nabla_{\mathbf{U}_{r,l}}\mathcal{L} = \Delta_{r,l}\mathbf{V}_{r,l}\left(\mathbf{S}_{r,l-1}\right)^{\top} \tag{21}$$

$$\nabla_{\mathbf{V}_{r_l}}\mathcal{L} = \left(\Delta_{r,l}\right)^{\top}\mathbf{U}_{r_l}\mathbf{S}_{r,l-1}. \tag{22}$$

For the first layer, i.e., $l = 1$, the expressions reduce to

$$\nabla_{\mathbf{U}_{r,1}}\mathcal{L} = \Delta_{r,1}\mathbf{V}_{r,1} \tag{23}$$

$$\nabla_{\mathbf{V}_{r,1}}\mathcal{L} = \left(\Delta_{r,1}\right)^{\top}\mathbf{U}_{r,1}. \tag{24}$$

The gradient with respect to the endmember $\mathbf{m}_r$ is

$$\nabla_{\mathbf{m}_r}\mathcal{L} = \mathbf{E}^{\top}\text{vec}\left(\mathbf{S}_r\right). \tag{25}$$

With the gradients, the parameters are updated using standard stochastic gradient descent

$$\mathbf{U}_{r,l} \leftarrow \mathbf{U}_{r,l} - \eta\nabla_{\mathbf{U}_{r,l}}\mathcal{L} \tag{26}$$

$$\mathbf{V}_{r,l} \leftarrow \mathbf{V}_{r,l} - \eta\nabla_{\mathbf{V}_{r,l}}\mathcal{L} \tag{27}$$

$$\mathbf{m}_r \leftarrow \mathbf{m}_r - \eta\nabla_{\mathbf{m}_r}\mathcal{L} \tag{28}$$

where η denotes the learning rate.

C. Implementation Details

We implemented ONTF using PyTorch. The depth of the nonlinear abundance decomposition was set to three. Specifically, the dimensions of $\mathbf{U}_{r,l}$ and $\mathbf{V}_{r,l}^{T}$ were progressively defined as $\{I \times I/2, J \times I/2\}$, $\{I/2 \times I/4, J/2 \times I/4\}$, and $\{I/4 \times I/8, J/4 \times I/8\}$. The total number of parameters is

$$\frac{21}{32}\left(W^2 + H^2\right)R + RB. \tag{29}$$

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In this way, the number of parameters exceeds the number of pixels, resulting in an overparameterized representation of the abundance map. The model was trained on an NVIDIA RTX 4090 GPU for 20 000 epochs with a learning rate of $4e^{-3}$. We opted for a fixed learning rate to establish a simple and robust baseline, thereby isolating the architectural contribution from the influence of advanced optimization schedules. The vertex component analysis (VCA) [52] algorithm was executed ten times, and the averaged endmembers were used to initialize $\mathbf{M}$. To avoid trivial solutions, values of $\mathbf{M}$ below zero were clipped to $1e^{-4}$.

IV. Experiments and Analysis

In this section, we conduct extensive experiments with alternative trackers to show the strong unmixing ability of our ONTF method.

A. Experiment Setup

1) Compared Methods: We compare our ONTF method with seven approaches, including the matrix-based unmixing method TV-RSNMF [53], the tensor-based unmixing method MV-NTF [22], and four deep learning-based unmixing methods: uDAS [33], UnDIP [54], DAEN [55], and A2SN [6]. Additionally, we include the model-driven deep learning method SNMF-Net [39] in the comparison. All parameters are set according to the recommendations in the original implementations.

2) Evaluation Indexes: To quantify the unmixing performance of each method, we measure the errors between the extracted endmembers and abundances against the reference ones. The spectral angle distance (SAD) calculates the spectral angle between the reference endmember $\mathbf{M}_r$ and the estimated endmember $\widehat{\mathbf{M}}_r$, defined as follows:

$$\mathrm{SAD}_r = \arccos\left(\frac{\mathbf{M}_r^\top \widehat{\mathbf{M}}_r}{\|\mathbf{M}_r\| \left\|\widehat{\mathbf{M}}_r\right\|}\right). \tag{30}$$

The root-mean-square error (RMSE) measures the difference between the reference abundance map $\mathbf{S}_r$ and the estimated abundance $\widehat{\mathbf{S}}_r$

$$\mathrm{RMSE}_r = \sqrt{\frac{1}{I \times J}\left\|\mathbf{S}_r - \widehat{\mathbf{S}}_r\right\|_F^2}. \tag{31}$$

3) Experimental Data: All methods were evaluated on both synthetic and real-world data. The real-world dataset includes Washington DC and Jasper Ridge HSIs. The synthetic data were generated following [23]. Six pure endmember signatures (Carnallite, Ammonio-jarosite, Almandine, Brucite, Axinite, and Chlorite) covering 224 spectral bands with wavelengths ranging from 0.38 to 2.5 μm were selected as endmembers. The signatures of the endmembers are shown in Fig. 2.

The abundance maps were produced as follows.

1) A synthetic image of size $z^2 \times z^2$ was partitioned into z^2 nonoverlapping regions of size $z \times z$.
2) Each region's pixels were filled with two randomly selected endmembers, mixed with ratios θ and $1 - \theta$.
3) A low-pass Gaussian filter with variance 2 was applied to each region.

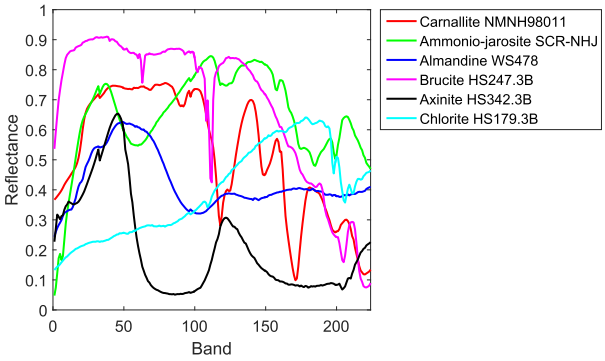


Fig. 2. Spectral signatures of six endmembers used in the synthetic data.

4) The abundance fractions of each pixel were rescaled to satisfy the ASC.

To test the sensitivity of each method to noise, the clean HSI was further corrupted with zero-mean additive Gaussian noise. The signal-to-noise ratio (SNR) is defined as follows:

$$\mathrm{SNR} = 10\log_{10}\frac{\mathbb{E}\left(\mathbf{x}^\top \mathbf{x}\right)}{\mathbb{E}\left(\mathbf{e}^\top \mathbf{e}\right)} \tag{32}$$

where $\mathbf{x}$ and $\mathbf{e}$ denote the clean signal and noise vectors at a pixel, respectively, and $\mathbb{E}(\cdot)$ denotes the expectation.

B. Experiments on Synthetic Data

1) Impact of Depth: Depth is the only hyperparameter that directly controls the performance of the proposed ONTF, where L denotes the network depth. A larger L corresponds to a deeper ONTF with more learnable parameters. In this experiment, synthetic data are generated with $z = 9$ and an SNR of 25 dB. By varying L from 1 to 5, the unmixing performance of ONTF is evaluated, as shown in Fig. 3.

As can be observed, the unmixing performance improves as L increases from 1 to 3. This improvement arises because a larger depth enables more effective hierarchical modeling of the low-rank structure in abundance maps, thereby capturing richer spatial dependencies. Moreover, the implicit regularization induced by gradient descent-based optimization further facilitates the recovery of intrinsic low-rank characteristics. However, when L becomes too large, performance degrades due to overparameterization, which increases optimization difficulty and reduces generalization ability. Based on these observations, $L = 3$ is adopted in all subsequent experiments.

To provide a more intuitive understanding of the low-rank estimation capability, Fig. 4 presents the estimated abundance maps and their corresponding singular value curves, along with the ground truth for comparison. It can be seen that ONTF accurately recovers the abundance distributions and yields singular value curves that closely match the ground truth. The main reason is that gradient descent-based optimization implicitly introduces regularization, which steers the solution toward low-dimensional feasible regions.

2) Comparison Under Different Mixing Levels: A higher mixing level corresponds to increased unmixing difficulty. In this experiment, the mixing coefficient θ is varied from 0.5 to 1.0 with a step size of 0.1 to investigate its impact on

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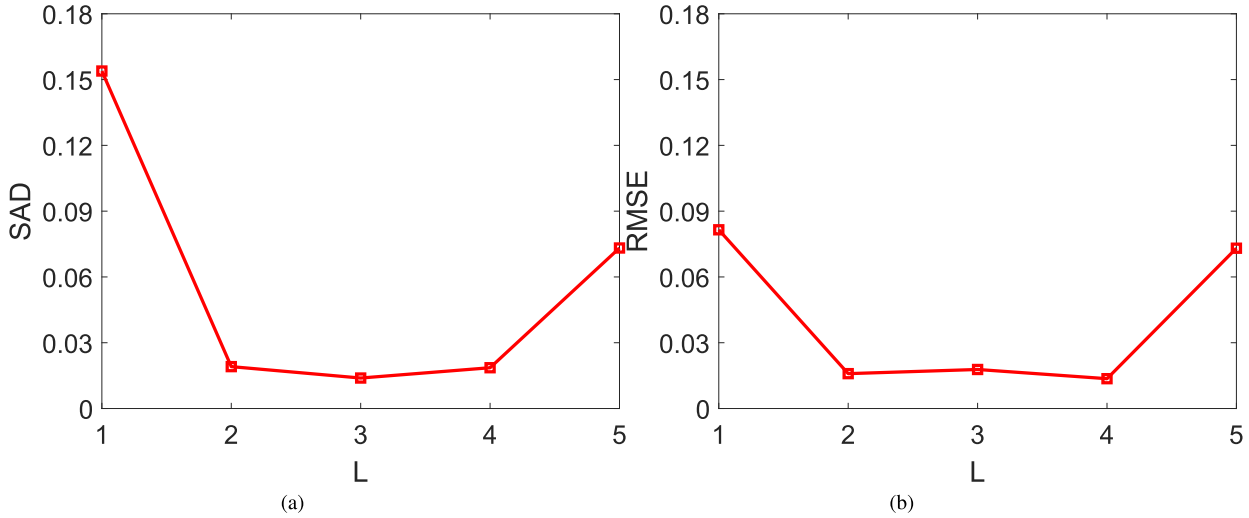


Fig. 3. Impact of depth in terms of (a) SAD and (b) RMSE.

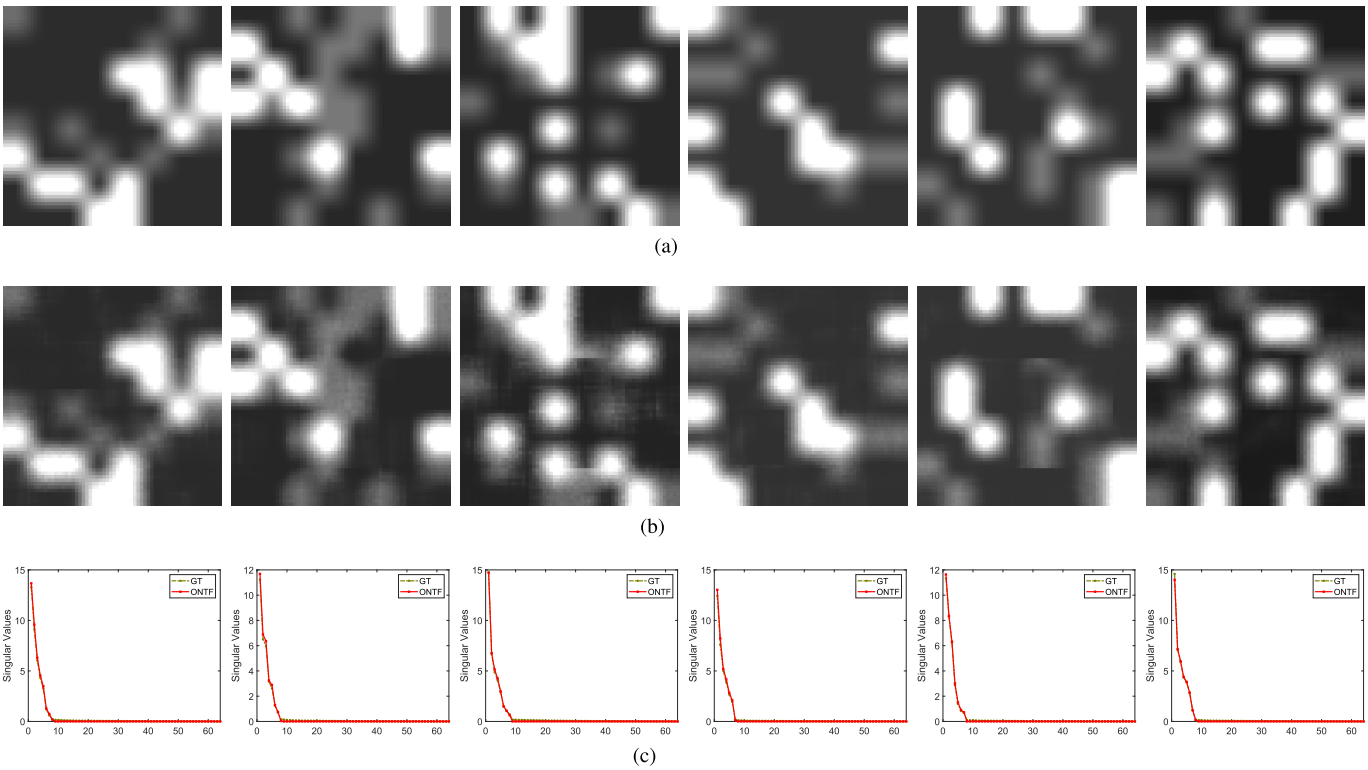


Fig. 4. Recovered abundances and their singular values. (a) GT. (b) ONTF. (c) SVD curves.

unmixing performance. Moreover, noise with 25 dB is added to the generated HSIs. MV-NTF is included as a representative baseline for comparison. As shown in Fig. 5, the proposed ONTF consistently outperforms MV-NTF across all mixing levels. Notably, this performance advantage becomes more pronounced as the mixing degree increases, indicating stronger robustness under more challenging mixing conditions. This superiority can be attributed to the gradient descent-based optimization strategy adopted by ONTF, which implicitly introduces regularization effects that more effectively enforce the intrinsic low-rank structure of the abundance maps, thereby enabling more accurate estimation of both endmembers and abundances.

3) Comparison Under Different Image Sizes: Larger images typically contain more complex spatial structures, which pose greater challenges for hyperspectral unmixing. In this experiment, we evaluate the impact of image size on unmixing performance by varying the spatial resolution from 25×25 to 36×36, 49×49, 64×64, and 100×100. Moreover, noise with 25 dB is added to the generated HSIs. As shown in Fig. 6, the proposed ONTF consistently achieves lower SAD and RMSE across all image sizes, demonstrating its strong and stable unmixing capability. These results indicate that ONTF effectively exploits low-rank structures and maintains robustness as spatial complexity increases.

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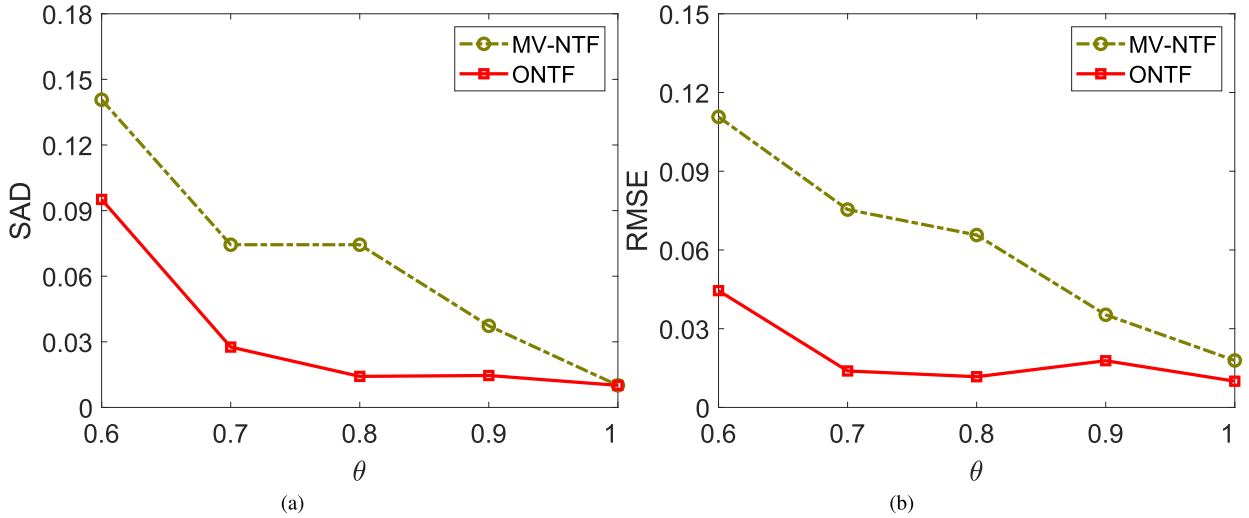


Fig. 5. Performance comparison with respect to different mixing levels in terms of (a) SAD and (b) RMSE.

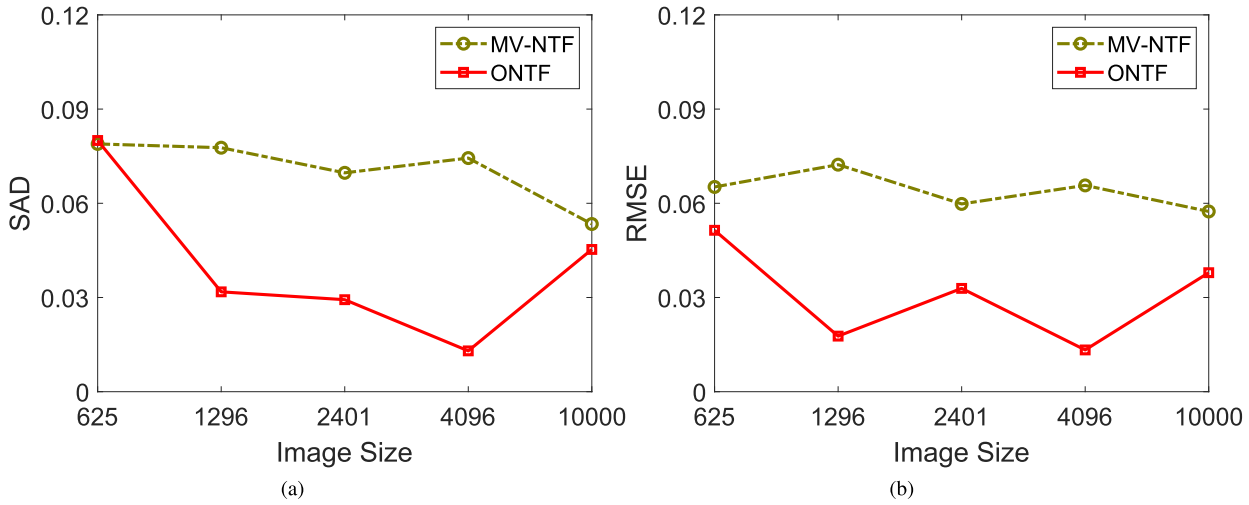


Fig. 6. Performance comparison with respect to different image sizes in terms of (a) SAD and (b) RMSE.

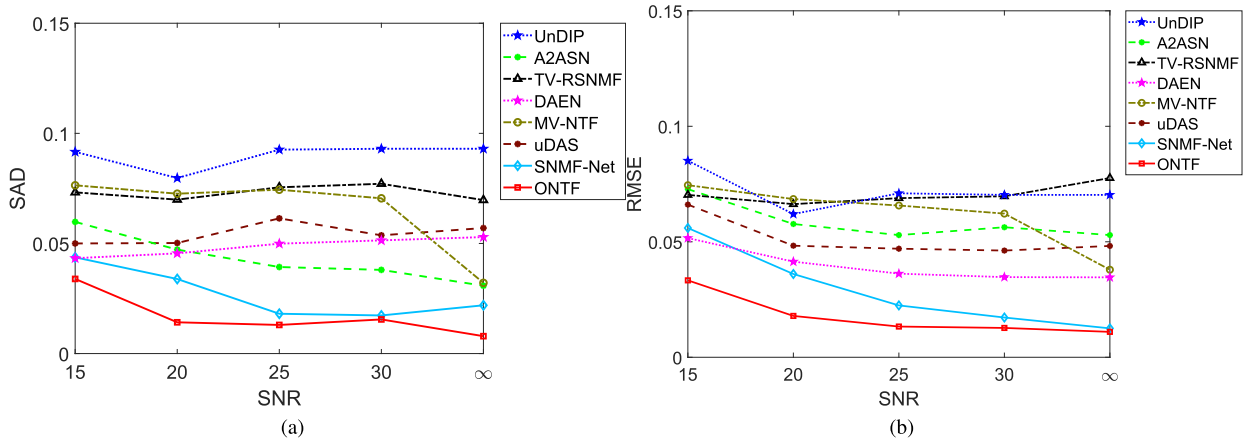


Fig. 7. Performance of all eight unmixing methods with respect to different noise levels in terms of (a) SAD and (b) RMSE.

4) Robustness to Noise: By varying the noise level from 15, 20, 25, and 30 dB to the noise-free case, the unmixing performance of all methods is reported in Fig. 7. As can be observed, MV-NTF consistently outperforms TV-RSNMF, which can be attributed to its inherent advantage in preserving spatial–spectral structures through tensor modeling. Compared with UnDIP, AEASN, uDAS, and DAEN, which adopt general deep architectures with relatively weak connections to the

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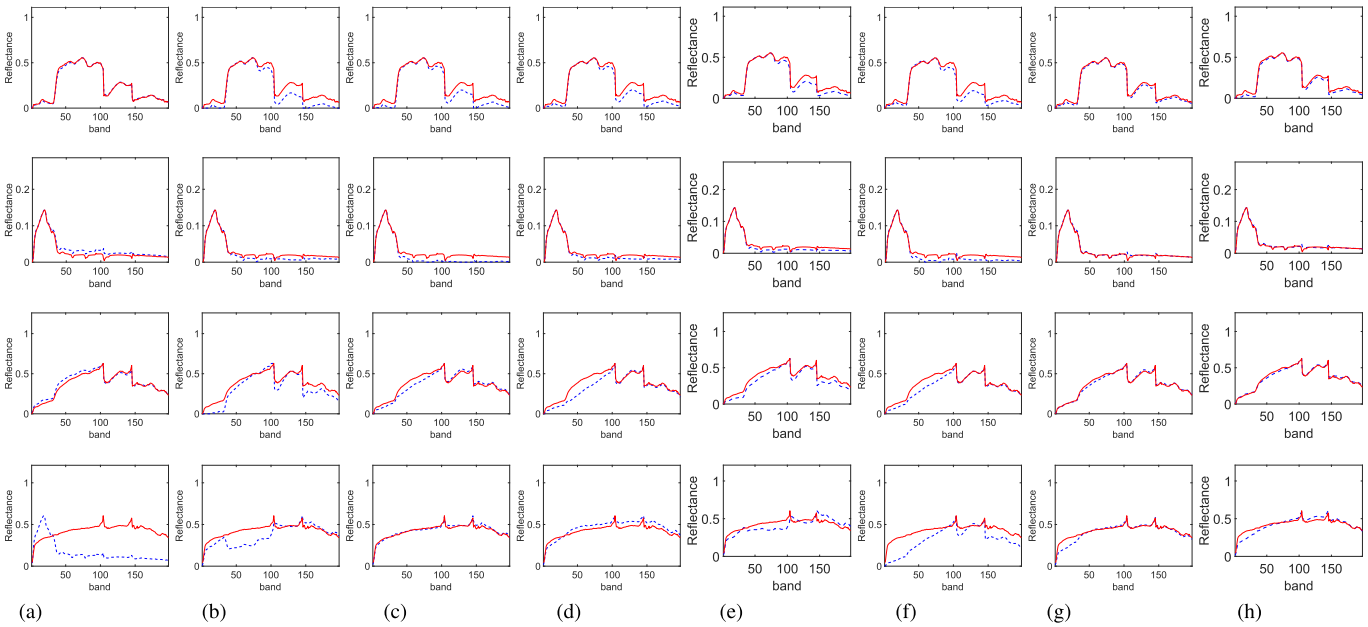


Fig. 8. Estimated endmembers by eight unmixing methods on the Jasper Ridge dataset. From top to bottom, the rows are the spectral signatures of tree, water, soil, and road. Solid lines denote the reference endmembers and dashed lines denote the estimated endmembers. (a) TV-RSNMF. (b) MV-NTF. (c) DAEN. (d) uDAS. (e) A2ASN. (f) UnDIP. (g) SNMF-Net. (h) ONTF.

TABLE I
MEANS AND STANDARD DEVIATIONS OF THE SAD ON JASPER RIDGE DATASET

Algorithm	TV-RSNMF [53]	MV-NTF [22]	DAEN [55]	uDAS [33]	A2ASN [6]	UnDIP [54]	SNMF-Net [39]	ONTF
Tree	0.0776±4.91%	0.2209±2.17%	0.1774±0.82%	0.1511±1.37%	0.1274±0%	0.1493±0%	**0.0478±1.62%**	0.0692±1.27%
Water	0.1337±1.31%	0.2352±1.65%	0.3237±1.42%	0.1742±11.39%	0.1627±0%	0.2529±0%	0.0855±1.77%	**0.0672±0.71%**
Soil	0.1047±10.11%	0.1752±4.67%	0.1123±2.03%	0.1639±6.09%	0.0957±0%	0.1144±0%	0.0443±3.09%	**0.0384±0.55%**
Road	0.6515±27.46%	0.1741±4.35%	**0.0588±1.59%**	0.0597±0.27%	0.1519±0%	0.2757±0%	0.0875±0.05%	0.0942±2.95%
Mean	0.2419±3.06%	0.2015±1.77%	0.1680±0.43%	0.1373±3.27%	0.1344±0%	0.1981±0%	**0.0663±1.04%**	0.0672±0.73%

underlying physical mixing model, SNMF-Net achieves better performance due to its tighter integration with the LMM and its ability to embed optimization-inspired priors into the network design. Among all competing methods, the proposed ONTF achieves the best robustness under different noise levels. This superior performance can be attributed to two main factors. First, ONTF explicitly exploits the hierarchical low-rank structure of abundance maps through nonlinear tensor factorization, resulting in a more expressive and noise-resilient unmixing model. Second, the gradient descent-based optimization framework enables more effective recovery of the intrinsic low-rank properties of both abundance maps and endmembers, particularly in noisy scenarios.

C. Experiments on Real-World Data

We further conduct an experiment on the real-world Jasper Ridge and Washington DC. In order to enable the quantitative comparison, we select a number of regions that most likely contain the pure pixels. The average spectrum of these regions is assigned as endmembers. After that, the CVX toolbox is used for abundance estimation.

1) Jasper Ridge Dataset: The Jasper Ridge dataset is one of the most widely used benchmark datasets for hyperspectral unmixing. It was acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) and consists of 512×614 pixels with 224 spectral bands. For experimental evaluation, a 100×100 subimage is cropped from the original scene. Following standard practice, bands heavily affected by water vapor absorption and atmospheric noise (bands 1–3, 108–112, 154–166, and 220–224) are removed, resulting in 198 bands for unmixing. The Jasper Ridge contains four dominant material classes, i.e., tree, water, soil, and road.

Table I reports the comparison in terms of the SAD metric. As can be observed, the proposed ONTF demonstrates superior capability in accurately extracting the water and soil endmembers. Moreover, even when initialized using VCA, ONTF achieves an average SAD that is comparable to that of SNMF-Net, which benefits from a more favorable initialization using endmembers extracted by the $L_{1/2}$-NMF [15] method. The spectral signatures of the extracted endmembers are illustrated in Fig. 8. It can be seen that ONTF produces spectral curves that most closely match the reference endmembers, further confirming its strong unmixing ability.

Fig. 9 presents the abundance maps estimated by all compared methods. As shown, the proposed ONTF is able to extract the tree and soil components with less noise. This improvement can be attributed to the hierarchical low-rank modeling strategy, which more effectively captures spatial

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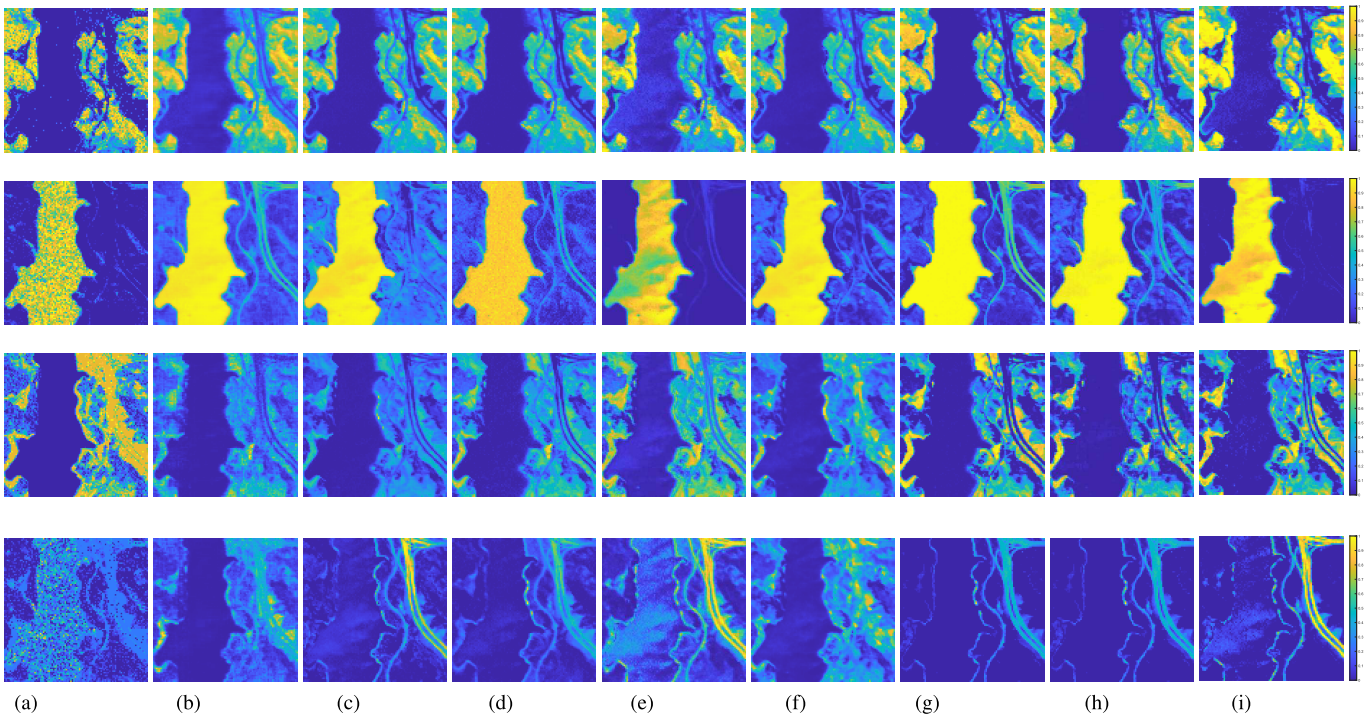


Fig. 9. Estimated abundance maps by eight unmixing methods on the Jasper Ridge dataset. From top to bottom, the rows are the abundance maps of tree, water, soil, and road. (a) TV-RSNMF. (b) MV-NTF. (c) DAEN. (d) uDAS. (e) A2ASN. (f) UnDIP. (g) SNMF-Net. (h) ONTF. (i) Reference.

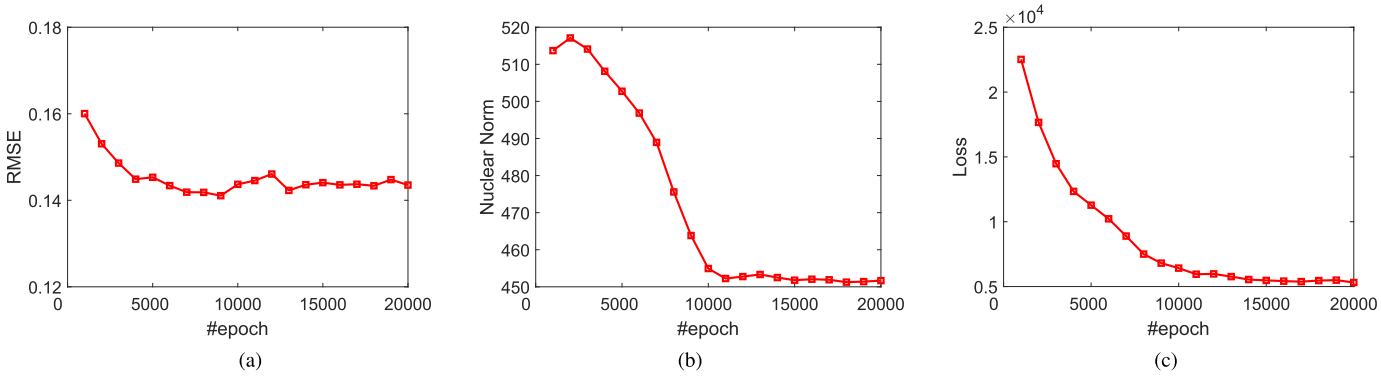


Fig. 10. Evolution of unmixing performance during training. (a) RMSE. (b) Nuclear norm. (c) Loss function.

correlations and suppresses noise interference. Overall, these results further demonstrate the strong unmixing capability of the proposed ONTF on real-world hyperspectral data.

To intuitively visualize the evolution of unmixing performance during the training process, we display the results at intervals of 1000 epochs. We employ the matrix nuclear norm as a proxy metric to quantify the low-rank property of the abundance maps. As illustrated in Fig. 10, both the RMSE and the nuclear norm decrease gradually over time. This phenomenon demonstrates that as the network trains, it effectively learns to exploit the underlying low-rank structure of the abundance maps, leading to progressively more accurate unmixing performance.

D. Washington DC Mall Dataset

The Washington DC Mall dataset was acquired by the Hyperspectral Digital Imagery Collection Experiment (HYDICE) sensor. Compared with the Jasper Ridge dataset, this dataset poses a more challenging hyperspectral unmixing task due to its complex urban environment and high spectral variability. The original image has a spatial size of 1208×307 pixels with a spatial resolution of approximately 2–3 m per pixel and contains 210 spectral bands covering the wavelength range from 0.4 to 2.4 μm. For our experiments, a subimage of size 150×150 pixels is cropped. Following standard preprocessing procedures, spectral bands affected by water absorption and severe noise are removed, resulting in 191 bands for unmixing. In this study, five primary materials are considered: tree, grass, street, roof, and water.

Table II reports the comparison results of all methods in terms of the SAD metric. As observed, the proposed ONTF achieves the most promising performance for the tree class and yields the best average SAD across all materials.

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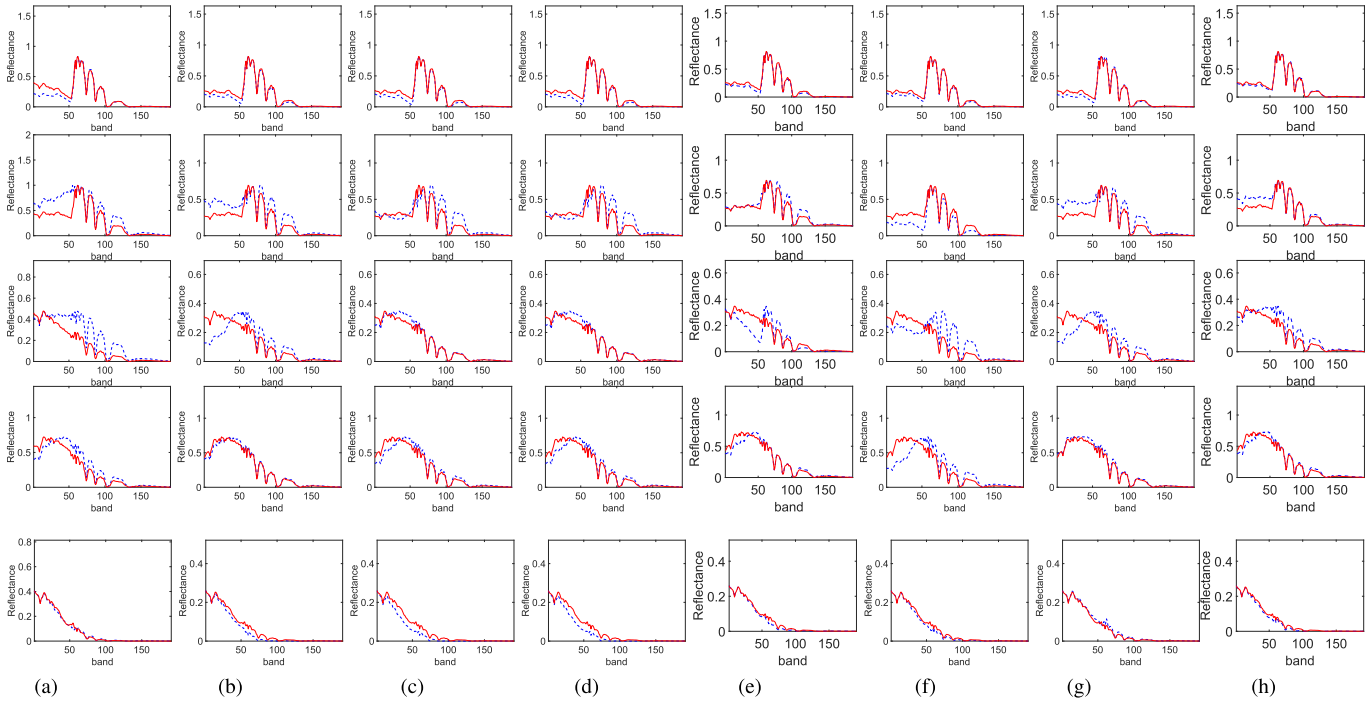


Fig. 11. Estimated endmembers by eight unmixing algorithms on the Washington DC mall data. From top to bottom, the rows are the spectral signatures of tree, grass, street, roof, and water. (a) TV-RSNMF. (b) MV-NTF. (c) DAEN. (d) uDAS. (e) A2ASN. (f) UnDIP. (g) SNMF-Net. (h) ONTF.

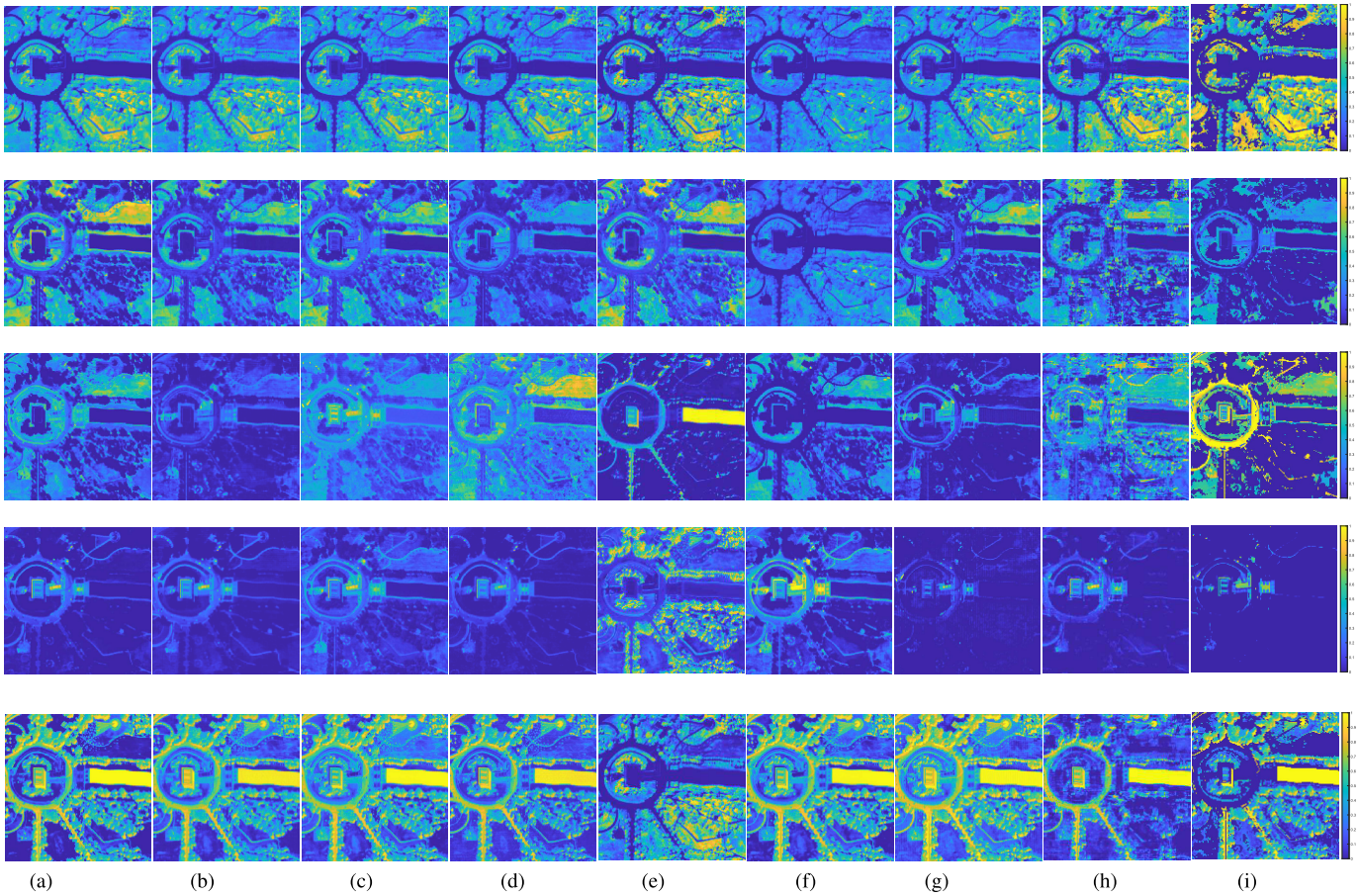


Fig. 12. Estimated abundance maps by eight unmixing algorithms on the Washington DC mall data. From top to bottom, the rows are the abundance maps of tree, grass, street, roof, and water. (a) TV-RSNMF. (b) MV-NTF. (c) DAEN. (d) uDAS. (e) A2ASN. (f) UnDIP. (g) SNMF-Net. (h) ONTF. (i) Reference.

Figs. 11 and 12 illustrate the extracted endmember signatures and abundance maps, respectively. Overall, the proposed ONTF demonstrates superior unmixing capability on this challenging urban dataset and produces results that more

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TABLE II
MEANS AND STANDARD DEVIATIONS OF THE SAD ON WASHINGTON DC MALL DATA

Algorithm	TV-RSNMF [53]	MV-NTF [22]	DAEN [55]	uDAS [33]	A2ASN [6]	UnDIP [54]	SNMF-Net [39]	ONTF
Tree	0.20711±1%	0.1211±0.53%	0.1378±2.33%	0.0939±2.15%	0.0887±0%	0.0795±0%	0.1309±1.06%	**0.0575±0.29%**
Grass	0.2961±4.34%	0.2357±0.40%	0.3773±17.84%	0.2888±6.03%	**0.1346±0%**	0.2483±0%	0.2180±2.09%	0.2209±4.87%
Street	0.3338±1.57%	0.4183±0.75%	0.3870±17.49%	0.3270±10.34%	0.3689±0%	0.4216±0%	0.3836±0.79%	**0.3224±3.20%**
Roof	0.2163±2.84%	0.0818±3.46%	0.1855±10.50%	0.0801±3.06%	0.1328±0%	0.3539±0%	**0.0356±0.51%**	0.1426 ±2.75%
Water	**0.0466±0.53%**	0.1317±0.70%	0.1760±5.88%	0.1264±9.28%	0.0835±0%	0.0784±0%	0.0602±1.70%	0.0625±2.07%
Mean	0.2200±0.51%	0.1977±0.80%	0.2527±3.42%	0.1833±1.99%	0.1617±0%	0.2363±0%	0.1657±0.63%	**0.1612 ±0.14%**

closely match the reference endmembers and abundance distributions.

V. CONCLUSION

In this article, we proposed an ONTF method for hyperspectral unmixing under the LMM framework. By introducing a hierarchical low-rank factorization, ONTF effectively captures the intrinsic low-rank structure of abundance maps, thereby constraining the solution space. Moreover, gradient descent-based optimization induces an implicit regularization that further improves the stability and robustness of the unmixing process. Extensive experiments on both synthetic and real-world hyperspectral datasets demonstrate that ONTF consistently achieves superior unmixing performance compared to state-of-the-art methods. In future work, we plan to extend the ONTF framework to non-LMMs to better address the complex spectral interactions encountered in real-world scenarios. Moreover, we acknowledge that providing rigorous theoretical guarantees for the stability of overparameterized tensor factorization remains a challenging open problem. We believe that recent advances in understanding implicit regularization, such as those presented in [56] and [57], may hold the key to explaining these properties, and we plan to investigate this in our future work.

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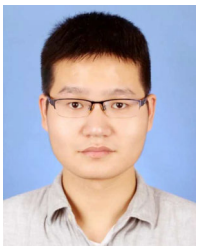
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